DUALIZING COMPLEXES ON A LOCALLY NOETHERIAN SCHEME DIFFER BY AN INVERTIBLE OBJECT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0ATP (sources/stacks-duality/pinned-original-source.tex line 547).

Allowed standard prerequisites (editorial).

- Uniqueness of dualizing complexes over a Noetherian ring. Dualizing Complexes, Lemma labelled dualizing-lemma-dualizing-unique, cited at proof lines 560-564. Gives invertibility and the evaluation isomorphism affine locally.

Source reference. Original lemma 0A7F: sources/stacks-chow/tags.txt, source lines 12878–12878. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 561 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let X be a locally Noetherian scheme. If K and K' are dualizing complexes on X, then K' is isomorphic to $K \otimes^{\mathbf{L}}_{\mathcal{O}_X} L$ for some invertible object L of $D(\mathcal{O}_X)$.*

Proof. Set

$$L = R\mathcal{H}om_{\mathcal{O}_X}(K, K')$$

This is an invertible object of $D(\mathcal{O}_X)$, because affine locally this is true, see Dualizing Complexes, Lemma 0A7F and its proof. The evaluation map $L \otimes^{\mathbf{L}}_{\mathcal{O}_X} K \to K'$ is an isomorphism for the same reason. □

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